Université Paris 1 Panthéon-Sorbonne
Master 2 Finance, Technology & Data

Final Exam Submission

Counterfactual Inflation Analysis

Ukraine in the Euro Area

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Course: Quantitative Methods in Finance

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1 Part A. Ukraine's Monetary Regime: A Preliminary Analysis (2000–2025)

1.1 Summary Table

Period	Approximate UAH/USD rate	De facto regime classification	Key events
Jan 2000 - Mar 2005	5.33	De facto peg (USD)	At the beginning of the year 2000, the NBU officially declared a free float but ended up regularly intervening in order to keep the exchange rate around UAH 5.33 per USD. Monetary policy at this time was used to keep the exchange rate stable and maintain the peg. (1)
Mar 2005 - Apr 2008	5.00–5.06	De facto peg (USD)	The NBU allowed the interbank exchange rate to fluctuate in a narrow range of UAH 5.00 - 5.06 per USD. However, the regime is still a de facto peg. (1)
Apr 2008 - Feb 2009	5.05 to 7.88	Managed exchange rate and peg breakdown	From March 2008, the NBU allowed the exchange rate to rise above the previous de facto range. Then the global financial crisis that occured during this time cause the peg to collapse. An IMF SBA (Stand By Arrangement) was approved in November 2008 and the IMF retroactively reclassified the regime from April 2008. (2)
Feb 2009 - Feb 2014	7.7 - 8.0	De facto peg (USD)	The exchange rate was effectively re-pegged to the US dollar but the crisis restrictions were not fully removed under the 2010 SBA. There IMF 2012 Article IV confirms a stablised arrangement from 1 March 2010. (3)(4)
Feb 2014 - Sep 2014	8.0 to 12–13	Forced float	Due to increasing pressure on the hryvnia and the critical reduction of the reserves, the NBU allowed the exchange rate to float in February 2014. Then in April 2014, the hryvnia depreciated by more than 35% due to the annexation of the Crimea. (5)(8)
Sep 2014 - Mar 2015	13 to 20–25	Managed float	This period is the second half of the crisis depreciation. Formally, the exchange rate remained floating but from September 2014 to March 2015 the NBU increased their crisis management. This was done through several price FX auctions, tighter intervention rules, a derivative ban and limitations on banks' FX positions. During 2014, the hryvnia fell by around 90%. Inflation reach 25% at the end of 2014 and 43% at the end of 2015. (6)(9)
Mar 2015 - Sep 2016	30.0 peak 23.5 end-2015 27.2 end-2016	Managed float with transition to inflation targeting	Following the exchange rate crisis, the NBU adopted a formally floating regime allowing the hryvnia to stabilise. Importantly, this period was still marked by extensive administrative FX restrictions. Policy focused on tight monetary control, rebuilding reserves and a gradual move toward inflation targeting. By mid 2016 the NBU progressively removed the emergency measures introduced in 2014 and 2015. (7)(8)(9)(10)

Period	Approximate UAH/USD rate	De facto regime classification	Key events
Sep 2016 - Feb 2019	27	Floating and inflation targeting	During this period, there was a focus on normalisation in inflation targeting. Despite the IMF classification of the exchange rate as floating from 2014 onward, capital controls and FX restrictions still remained. The NBU formally adopted inflation targeting in December 2016 and announced a declining target path from 12% in 2016 to 5% from 2019 onward. Over this period, the exchange rate was described as floating and intervention was limited to smoothing excess volatility and reubuilding reserves, not defending a peg. Anti-crisis FX restrictions were progressively rolled back. This was the most evident period of monetary autonomy in the sample. (10)(11)(12)(22)(23)
Feb 2019 - Dec 2021	Mid/High 20s to 27.3 end-2021	Floating and inflation targeting	Following the new FX regime which took effect on 7th February 2019, the NBU maintained a floating exchange rate within the inflation targeting framework. There is a continued unwinding of the 2014 and 2015 anti-crisis restrictions and the FX liberalisation continued through 2021. This included easing rules on FX purchases, derivatives and individuals foreign investments. In this period the exchange rate remained market-determined and interventions were again only used to smooth volatility and build reserves. (13)(21)(22)(23)
Dec 2021 - Oct 2023	29.25 - fixed at start of the invasion 36.57 - adjusted on 21 July 2022	Wartime fixed exchange rate	After the full-scale invasion of Ukraine, a wartime emergency exchange rate was introduced. Then on 24 February 2022, the NBU, aiming to preserve reserves, fixed the official exchange rate, limited cash withdrawals, maintained cashless payments, imposed FX restrictions and capital controls. This also aided in containing panic. In July 2022, after major wartime changes to economic fundamentals, the NBU devalued the official rate by 25% to 36.57. This was not a return to peacetime inflation targeting or to the market float but it was a wartime stabilisation regime. This was a based on a fixed nominal anchor and extensive administrative controls aimed at keeping the financial system functioning under martial law. At the end of the period, the NBU moved to managed exchange rate flexibility. (14)(15)(16)(17)

Period	Approximate UAH/USD rate	De facto regime classification	Key events
Oct 2023 - Dec 2025	42.04	Managed float and flexible inflation targeting	On the 3rd of October 2023, the NBU ended the wartime fixed-rate regime and shifted to managed flexibility. From then on, the official exchange rate was determined by interbank transactions rather than being set administratively. The IMF reclassified the de facto arrangement from stabilized to floating with effect from that 3rd October. In 2024, the NBU stated that it had transitioned to flexible inflation targeting as an intermediate regime as they worked their way to full-fledged inflation targeting with a floating exchange rate. The exchange rate was allowed to move in both directions but the NBU continued to cover the structural FX deficit, smooth excess volatility, and preserve reserves. By 2025 the NBU officially described the current framework as flexible inflation targeting supported by managed flexibility and still-existing FX restrictions. (18)(19)(20)

Note: Citations are presented in brackets and correspond to entries in the references section. These citations pertain to both approximate exchange rate and key events columns.

1.2 Brief Argument

During most of this period, Ukraine did not have a genuine monetary policy sovereignty. True sovereignty is seen only in two windows, being the post-2016 inflation-targeting period from September 2016 to February 2022, and the post-October 2023 period. From September 2016 to December 2021, the NBU formally adopted inflation targeting with a floating exchange rate and used interest rates for domestic stabilisation rather than to defend a peg. In the post-October 2023 period, the NBU engaged in managed flexibility, but this remained constrained by ongoing FX restrictions and wartime conditions. By contrast, from January 2000 to April 2008 and again from February 2009 to February 2014, monetary policy was de facto subordinated to maintaining the USD peg, so the NBU could not engage in independent demand management. The 2014-2016 crisis period involved some exchange-rate flexibility, but policy was dominated by crisis management rather than autonomous stabilisation. The wartime fixed-rate regime from February 2022 to October 2023 similarly suspended monetary independence. These important periodic distinctions imply that Euro Area membership would not reflect the same treatment for the whole sample. During peg periods, joining the euro would not have indicated a large additional loss of sovereignty, though some credibility benefit from importing the ECB's nominal anchor would still have been present given the NBU's weak institutional credibility before 2016. The treatment effect would be largest where Ukraine had genuine monetary sovereignty, such as during the post-2016 inflation-targeting period, and smallest during the long peg episodes where the NBU had already surrendered their independent monetary policy to an exchange-rate anchor.

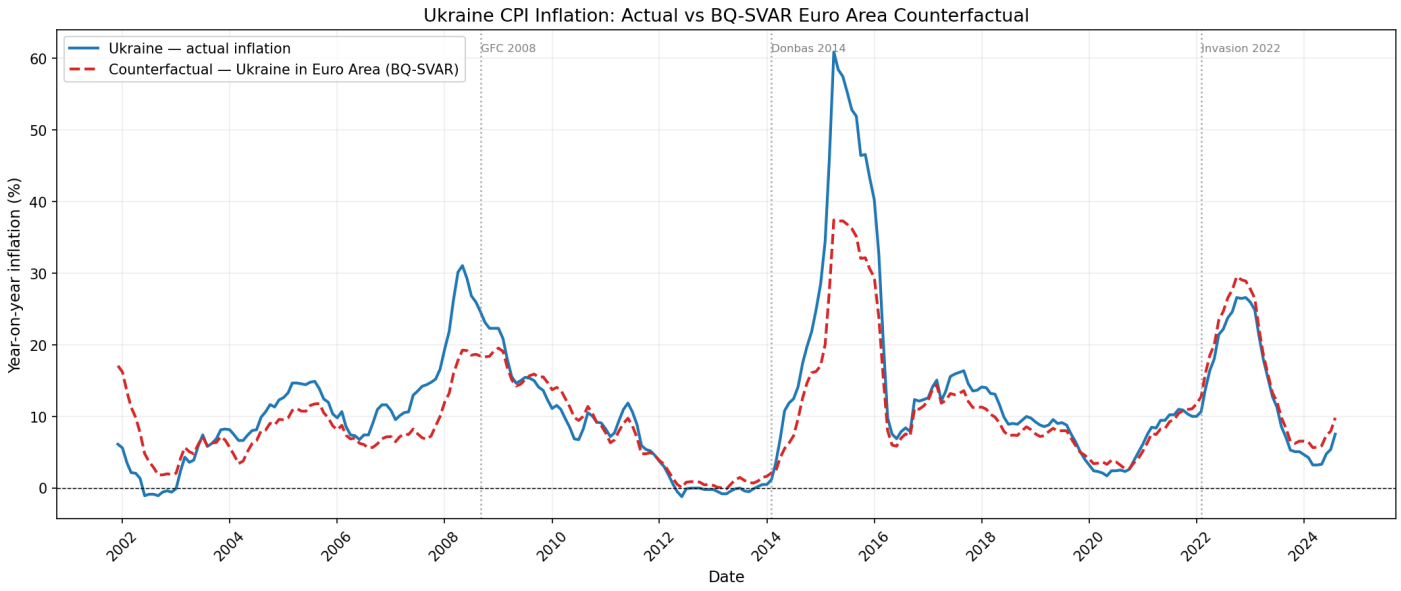

Figure 1: BQ-SVAR Ukraine in Euro Area Counterfactual. Rather than averaging euro area inflation, this approach feeds EA demand shocks through Ukraine's estimated VAR, preserving country-specific transmission.

2 Part B. Counterfactual Inflation Analysis

The BQ-SVAR counterfactual implies that the costs and benefits of monetary sovereignty changed considerably over the three crisis periods. In the 2008 to 2009 period, counterfactual inflation is lower than Ukraine's actual inflation at the crisis peak, but the gap remains much smaller than in 2014 to 2015. This suggests that Euro Area membership would have mitigated inflation during the global financial crisis, but that the inflation effects of monetary sovereignty were still more limited than during the later currency collapse. As Ukraine was already under a hard-dollar peg, it lacked effective monetary sovereignty, so Euro Area membership would have had a more limited additional impact on inflation dynamics than in 2014 to 2015. In the 2014 to 2015 period, actual inflation stood at 60%, while counterfactual inflation was around 29%. This difference reflects the collapse of the hryvnia, which pushed domestic prices to rise sharply through import cost inflation. This channel would have been largely absent if Ukraine had adopted the euro, since the exchange rate would have been fixed. Thus, Euro Area membership would have prevented this devaluation-driven inflation spike. In the 2022 crisis, counterfactual inflation rises above actual inflation, meaning that under Euro Area membership, external shocks, especially the European energy shock, would have passed more directly into Ukrainian inflation. However, under Euro Area membership, Ukraine would have lacked the autonomy to offset this through exchange-rate and liquidity policies. Overall, the counterfactual does not simply imply that Ukraine would have experienced lower inflation had it been inside the Euro Area. Without its own central bank to provide emergency liquidity, Ukraine would have lost an important institutional tool on which it relied under martial law. Thus, it might have avoided a currency crisis while becoming more exposed to financing and debt-sustainability pressures in the future. The true cost of monetary sovereignty depends on the kind of macroeconomic crisis to which a country is most exposed. For Ukraine, retaining monetary sovereignty meant absorbing inflation spikes caused by devaluation, while preserving the emergency policy flexibility that became crucial in 2022.

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